

Mohamad Nouval Amrullah

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PROFESSIONAL SUMMARY

Final-year (8th semester) Computer Engineering Technology student with hands-on experience in IoT systems, backend development, and information security. Experienced in building real-time monitoring systems, integrating industrial devices, and developing data-driven web applications for visualization and analysis.

Proven ability to design end-to-end systems, from data acquisition and communication to backend processing and user-facing dashboards. Also experienced in automating security-related processes, supporting compliance activities, and monitoring system anomalies.

Strong problem-solving mindset with a focus on efficiency, scalability, and real-world impact. Interested in leveraging technology to support digital transformation and improve operational performance.

EDUCATION

Bachelor in Computer Engineering, College of Vocational Studies of IPB University

Aug '22 – Now

- GPA: 3.53 / 4.00 (Temporary)
- Relevant Courework: mobile application, web progamming, network application administration, software engineer, hardware engineer, Information Security.

EXPERIENCES

Information Security & Compliance Intern – (Mekari)

Oct'25-Apr'26

- Automated internal security processes and improved existing workflows within the Infosec & Compliance team.
- Conducted ISMS onboarding training, vendor security reviews, and managed secure data deletion for offboarded users.
- Monitored anomalous login activities and contributed to phishing awareness campaigns.
- Developed and documented data deletion procedures in compliance with internal SOP.

Operational Excellent Intern – (PT URC Indonesia)

Aug'25-Feb'26

- Designed and implemented an IoT-based monitoring system integrating industrial PLCs with real-time web dashboards.
- Built end-to-end data pipelines for machine data acquisition, processing, and visualization.
- Developed real-time monitoring with historical tracking, enabling production traceability and basic MES functionality.
- Implemented OEE calculation and improved operational visibility through data-driven insights.

Freelance Full Stack Developer – (Custom Sales & Invoicing System) for Traditional Market Business

Mei-Jun'25

- Developed a full-stack web system for managing pricing, orders, delivery, and invoicing for a real-world client.
- Designed and implemented a normalized relational database schema to support complex business workflows.
- Built dynamic multi-step forms and implemented business logic for order processing, partial deliveries, and invoicing.
- Deployed features such as PDF generation and user dashboards; system is actively used in daily operations. The system is now actively used by the client for their daily business operations.

Internship Backend Developer, (PT. Aero System Indonesia)

Feb–Mei '25

- I am currently an intern working on developing a real-time monitoring system using the ELK (Elasticsearch, Logstash, Kibana) stack and Fluent-bit to implement a log service running on Docker. My main tasks involve utilizing Elasticsearch and Logstash, parsing data via URL/port from Elasticsearch into a Laravel 10 project. The goal of this implementation is to display real-time log service data efficiently.

PKL, (Depohar 10 Sathar 15)

Jan–Mar '22

- Aircraft Engine Cable Troubleshooting identifying, analyzing, and resolving issues within the electrical wiring system of aircraft engines. This includes inspecting cable connections, performing measurements with specialized tools, and ensuring the system operates optimally according to aviation standards.

MAKING PROJECT EXPERIENCES

(Student), (Computer Engineering)

Aug '22 – Now

Software Engineering – Application and Website:**Website Development:**

- Built various applications and websites using Java, Laravel, React.js, Next.js, and PHP.
- One of the key projects involved developing an employee information system using Laravel, including features such as authentication, data management, and a real-time dashboard.
- Implemented RESTful APIs, user authentication, and MySQL database integration.

Aquarium Water Pump Automation System:

- Designed an automatic water pump control system for aquariums using water level sensors and microcontrollers.
- Implemented control logic using a GAL16V8 IC and created the schematic and PCB layout.
- Successfully built an automated system that activates the pump based on water levels, optimizing energy efficiency.

Clock Project:

- Developed a precise digital clock system using ICs and Arduino-based microcontrollers.
- Programmed real-time clock display via 7-segment displays and designed the PCB layout using Eagle.

HONORS & AWARDS

(Runner up) of (In The Internasional Final Paper Competition Of Circular Economy), (College Of Vocational Studies 2023)

Jul '23

- Role in the Project

As part of the winning team, I was responsible for hardware development to support the implementation of Circular Economy in sustainable agricultural products.

Main Contributions:

Designing and assembling schematic and hardware layouts. Utilizing tools such as microcontroller.

SKILLS

Hard Skills :

- **Programming & Web Development :** JavaScript, Java, PHP, HTML, CSS, React.js, Next.js, Laravel, CodeIgniter.
- **Data Analysis & Machine Learning :** Python, NumPy, Pandas, Matplotlib, Google Colab, Jupyter Notebook.
- **Database Management :** MongoDB, MySQL, PostgreSQL, Supabase.
- **DevOps & Tools :** Git, GitHub, Docker.
- **Hardware & IoT :** Arduino, Proteus, Eagle, Fritzing, PLC Omron NJ301, PLC Mitsubishi Fx3u, Siemens IOT2040, Sysmac Studio, Gx Works 2, NB Designer, Node-RED.
- **Networking :** Cisco networking equipment, TCP/IP, DNS, DHCP, VLAN, and basic routing and switching.
- **Productivity Tools:** Microsoft Word, Microsoft Excel.

Soft Skills :

- Strong communication and collaboration, Creative problem-solving, Critical and analytical thinking, Adaptability and time management under pressure, Teamwork across disciplines.